Two-Way Frequency Tables & Conditional Probability

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Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If one adult is selected at random from the group, what is the probability that the adult is in the category Southside and Support?

	Support	Oppose	Total
Northside	13	24	37
Southside	10	49	59
Total	23	73	96

2 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. If one student is selected at random from the group, what is the probability that the student is in the category Grade 11 and Morning?

	Morning	Evening	Total
Grade 10	24	20	44
Grade 11	9	53	62
Total	33	73	106

3 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If one book is selected at random from the group, what is the probability that the book is in the category Fiction and Digital?

	Print	Digital	Total
Fiction	36	55	91
Nonfiction	10	29	39
Total	46	84	130

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4 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If one commuter is selected at random from the group, what is the probability that the commuter is in the category Under 30 and Car?

	Public trans	Car	Total
Under 30	12	30	42
30 or older	39	47	86
Total	51	77	128

5 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey of 178 employees found that 12 are in the Marketing department and working remotely, 118 are in the Marketing department and working on-site, 35 are not in the Marketing department and working remotely, and 13 are not in the Marketing department and working on-site. What is the probability that a randomly selected employee is in the Marketing department?

6 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bag contains 5 red marbles and 15 blue marbles. If one marble is drawn at random, what is the probability that it is red?

7 [Foundational] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey of 178 shoppers found that 47 are using a coupon and spending over USD 50, 75 are using a coupon and spending USD 50 or less, 25 are not using a coupon and spending over USD 50, and 31 are not using a coupon and spending USD 50 or less. What is the probability that a randomly selected shopper is using a coupon?

Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

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8 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If a patient in the category Treatment is selected at random, what is the probability that the patient is also in the category Improved?

	Improved	Did not impro	Total
Treatment	11	38	49
Placebo	35	41	76
Total	46	79	125

9 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If a book in the category Nonfiction is selected at random, what is the probability that the book is also in the category Print?

	Print	Digital	Total
Fiction	52	38	90
Nonfiction	23	58	81
Total	75	96	171

10 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Northside is selected at random, what is the probability that the adult is also in the category Support?

	Support	Oppose	Total
Northside	50	32	82
Southside	14	51	65
Total	64	83	147

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11 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If a book in the category Fiction is selected at random, what is the probability that the book is also in the category Print?

	Print	Digital	Total
Fiction	21	55	76
Nonfiction	34	57	91
Total	55	112	167

12 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If a frog in the category Pond A is selected at random, what is the probability that the frog is also in the category Brown?

	Green	Brown	Total
Pond A	39	17	56
Pond B	17	54	71
Total	56	71	127

13 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If a patient in the category Placebo is selected at random, what is the probability that the patient is also in the category Did not improve?

	Improved	Did not impro	Total
Treatment	44	8	52
Placebo	17	22	39
Total	61	30	91

14 [Intermediate] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey of 207 voters found that 81 are under 40 years old and supporting the proposal, 5 are under 40 years old and opposing the proposal, 57 are 40 years old or older and supporting the proposal, and 64 are 40 years old or older and opposing the proposal. Given that a voter is under 40 years old, what is the probability that they are also supporting the proposal?

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Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

15 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. A student will be selected at random from those in the category Evening. What is the probability that the student is in the category Grade 11?

	Morning	Evening	Total
Grade 10	51	26	77
Grade 11	50	32	82
Total	101	58	159

16 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. A frog will be selected at random from those in the category Brown. What is the probability that the frog is in the category Pond B?

	Green	Brown	Total
Pond A	12	33	45
Pond B	55	44	99
Total	67	77	144

17 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. One entry in the table is missing. If a patient in the category Placebo is selected at random, what is the probability that the patient is in the category Improved?

	Improved	Did not impro	Total
Treatment	51	49	100
Placebo	?	38	96
Total	109	87	196

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18 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. A patient will be selected at random from those in the category Improved. What is the probability that the patient is in the category Treatment?

	Improved	Did not impro	Total
Treatment	48	58	106
Placebo	32	28	60
Total	80	86	166

19 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. One entry in the table is missing. If a adult in the category Southside is selected at random, what is the probability that the adult is in the category Oppose?

	Support	Oppose	Total
Northside	11	40	51
Southside	28	?	69
Total	39	81	120

20 [Challenging] • [SPR] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. A adult will be selected at random from those in the category Support. What is the probability that the adult is in the category Southside?

	Support	Oppose	Total
Northside	22	17	39
Southside	48	47	95
Total	70	64	134

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

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21 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If one commuter is selected at random from the group, what is the probability that the commuter is in the category Under 30 and Public transit?

	Public trans	Car	Total
Under 30	25	35	60
30 or older	52	33	85
Total	77	68	145

a. $\frac{5}{29}$

b. $\frac{5}{12}$

c. $\frac{25}{77}$

d. $\frac{12}{29}$

22 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If one frog is selected at random from the group, what is the probability that the frog is in the category Pond A and Brown?

	Green	Brown	Total
Pond A	26	50	76
Pond B	30	45	75
Total	56	95	151

a. $\frac{25}{38}$

b. $\frac{10}{19}$

c. $\frac{76}{151}$

d. $\frac{50}{151}$

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23 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If one frog is selected at random from the group, what is the probability that the frog is in the category Pond B and Brown?

	Green	Brown	Total
Pond A	38	43	81
Pond B	47	18	65
Total	85	61	146

a. $\frac{65}{146}$

b. $\frac{9}{73}$

c. $\frac{18}{65}$

d. $\frac{18}{61}$

24 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If one patient is selected at random from the group, what is the probability that the patient is in the category Treatment and Did not improve?

	Improved	Did not impro	Total
Treatment	47	10	57
Placebo	44	39	83
Total	91	49	140

a. $\frac{10}{57}$

b. $\frac{10}{49}$

c. $\frac{57}{140}$

d. $\frac{1}{14}$

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25 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. If one student is selected at random from the group, what is the probability that the student is in the category Grade 11 and Morning?

	Morning	Evening	Total
Grade 10	10	49	59
Grade 11	48	44	92
Total	58	93	151

a. $\frac{48}{151}$

b. $\frac{12}{23}$

c. $\frac{24}{29}$

d. $\frac{92}{151}$

26 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If one frog is selected at random from the group, what is the probability that the frog is in the category Pond A and Green?

	Green	Brown	Total
Pond A	46	8	54
Pond B	30	27	57
Total	76	35	111

a. $\frac{18}{37}$

b. $\frac{46}{111}$

c. $\frac{23}{27}$

d. $\frac{23}{38}$

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27 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If one patient is selected at random from the group, what is the probability that the patient is in the category Placebo and Improved?

	Improved	Did not impro	Total
Treatment	41	51	92
Placebo	36	47	83
Total	77	98	175

a. $\frac{83}{175}$

b. $\frac{36}{175}$

c. $\frac{36}{83}$

d. $\frac{36}{77}$

28 [Foundational] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If one book is selected at random from the group, what is the probability that the book is in the category Nonfiction and Digital?

	Print	Digital	Total
Fiction	48	14	62
Nonfiction	30	12	42
Total	78	26	104

a. $\frac{2}{7}$

b. $\frac{6}{13}$

c. $\frac{21}{52}$

d. $\frac{3}{26}$

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

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29 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If a commuter in the category 30 or older is selected at random, what is the probability that the commuter is also in the category Car?

	Public trans	Car	Total
Under 30	49	41	90
30 or older	34	27	61
Total	83	68	151

a. $\frac{34}{61}$

b. $\frac{27}{61}$

c. $\frac{27}{151}$

d. $\frac{27}{68}$

30 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Southside is selected at random, what is the probability that the adult is also in the category Oppose?

	Support	Oppose	Total
Northside	40	48	88
Southside	55	59	114
Total	95	107	202

a. $\frac{59}{202}$

b. $\frac{59}{107}$

c. $\frac{55}{114}$

d. $\frac{59}{114}$

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31 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Northside is selected at random, what is the probability that the adult is also in the category Oppose?

	Support	Oppose	Total
Northside	20	34	54
Southside	27	46	73
Total	47	80	127

a. $\frac{17}{27}$

b. $\frac{34}{127}$

c. $\frac{17}{40}$

d. $\frac{10}{27}$

32 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. If a student in the category Grade 11 is selected at random, what is the probability that the student is also in the category Morning?

	Morning	Evening	Total
Grade 10	9	35	44
Grade 11	27	13	40
Total	36	48	84

a. $\frac{3}{4}$

b. $\frac{13}{40}$

c. $\frac{27}{40}$

d. $\frac{9}{28}$

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33 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. If a book in the category Fiction is selected at random, what is the probability that the book is also in the category Print?

	Print	Digital	Total
Fiction	9	17	26
Nonfiction	29	54	83
Total	38	71	109

a. $\frac{9}{26}$

b. $\frac{9}{109}$

c. $\frac{9}{38}$

d. $\frac{17}{26}$

34 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A city surveyed commuters about how they travel to work, by age group. The results are shown in the table. If a commuter in the category Under 30 is selected at random, what is the probability that the commuter is also in the category Public transit?

	Public trans	Car	Total
Under 30	33	30	63
30 or older	49	34	83
Total	82	64	146

a. $\frac{33}{146}$

b. $\frac{33}{82}$

c. $\frac{10}{21}$

d. $\frac{11}{21}$

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35 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. If a adult in the category Southside is selected at random, what is the probability that the adult is also in the category Oppose?

	Support	Oppose	Total
Northside	12	46	58
Southside	23	48	71
Total	35	94	129

a. $\frac{16}{43}$

b. $\frac{24}{47}$

c. $\frac{23}{71}$

d. $\frac{48}{71}$

36 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. If a frog in the category Pond B is selected at random, what is the probability that the frog is also in the category Brown?

	Green	Brown	Total
Pond A	10	42	52
Pond B	15	11	26
Total	25	53	78

a. $\frac{11}{78}$

b. $\frac{11}{53}$

c. $\frac{15}{26}$

d. $\frac{11}{26}$

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37 [Intermediate] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A clinic recorded whether patients in a study received a new treatment and whether their symptoms improved. The results are shown in the table. If a patient in the category Placebo is selected at random, what is the probability that the patient is also in the category Did not improve?

	Improved	Did not impro	Total
Treatment	24	32	56
Placebo	17	22	39
Total	41	54	95

a. $\frac{22}{95}$

b. $\frac{11}{27}$

c. $\frac{17}{39}$

d. $\frac{22}{39}$

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

38 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. One entry in the table is missing. If a student in the category Grade 10 is selected at random, what is the probability that the student is in the category Evening?

	Morning	Evening	Total
Grade 10	44	?	93
Grade 11	14	43	57
Total	58	92	150

a. $\frac{44}{93}$

b. $\frac{49}{150}$

c. $\frac{49}{92}$

d. $\frac{49}{93}$

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39 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. A student will be selected at random from those in the category Morning. What is the probability that the student is in the category Grade 10?

	Morning	Evening	Total
Grade 10	36	35	71
Grade 11	36	34	70
Total	72	69	141

a. $\frac{36}{71}$

b. $\frac{12}{47}$

c. 1

d. $\frac{1}{2}$

40 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. For which group is the proportion in the category Green greater?

	Green	Brown	Total
Pond A	48	55	103
Pond B	26	45	71
Total	74	100	174

a. Pond A

b. Pond B

c. The proportions are equal.

d. There is not enough information to compare.

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41 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A biologist classified the frogs captured at two ponds by color. The results are shown in the table. One entry in the table is missing. If a frog in the category Pond B is selected at random, what is the probability that the frog is in the category Green?

	Green	Brown	Total
Pond A	48	14	62
Pond B	?	23	62
Total	87	37	124

a. $\frac{23}{62}$

b. $\frac{39}{124}$

c. $\frac{13}{29}$

d. $\frac{39}{62}$

42 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. For which group is the proportion in the category Print greater?

	Print	Digital	Total
Fiction	53	29	82
Nonfiction	10	27	37
Total	63	56	119

a. Fiction

b. Nonfiction

c. The proportions are equal.

d. There is not enough information to compare.

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43 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A bookstore recorded the format of the books it sold in one week, by genre. The results are shown in the table. One entry in the table is missing. If a book in the category Nonfiction is selected at random, what is the probability that the book is in the category Digital?

	Print	Digital	Total
Fiction	45	26	71
Nonfiction	51	?	101
Total	96	76	172

a. $\frac{50}{101}$

b. $\frac{51}{101}$

c. $\frac{25}{86}$

d. $\frac{25}{38}$

44 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A group of students was asked whether they prefer to study in the morning or in the evening. The results are shown in the table, by grade level. For which group is the proportion in the category Morning greater?

	Morning	Evening	Total
Grade 10	35	34	69
Grade 11	27	56	83
Total	62	90	152

a. Grade 11

b. The proportions are equal.

c. There is not enough information to compare.

d. Grade 10

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45 [Challenging] • [MCQ] • Target: Two-Way Frequency Tables, Joint & Conditional Probability

A survey asked adults in two neighborhoods whether they support building a new park. The results are shown in the table. For which group is the proportion in the category Support greater?

	Support	Oppose	Total
Northside	30	52	82
Southside	39	54	93
Total	69	106	175

a. There is not enough information to compare.

b. Southside

c. Northside

d. The proportions are equal.

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